Tutorial 2: Optimisation model

This tutorial trains you to quantify the elements of an optimisation model:

- **Decision variables**: quantities of resources used (e.g., money allocated) or levels of activities (e.g., units produced, gallons blended).
- **Objective**: a function of decision variables to be maximised/minimised.
- **Constraints**: relationships/limits the decision variables must satisfy.

Example 1: Box design (maximise volume)

With precisely 2700 cm^2 of cardboard, construct a box with width x, depth y, height z and volume $V = xyz$. The width must be double the depth.

Decision variables

Objective

Constraints

1. Surface area constraint:

2. Width-depth relation:

Lagrangian set-up A Lagrangian expression is written (including nonnegativity multipliers) as

with first-order conditions :

and sign restrictions $x, y, z > 0$ and $\lambda_3, \lambda_4, \lambda_5 > 0$.

Example 2: Wood panelling production

A manufacturing facility produces four types of wood panelling: Tahoe, Pacific, Savannah, and Aspen. Each product is manufactured using pine chips and oak chips and requires both gluing and pressing processes.

The profit earned per pallet is as follows:

- \$450 for Tahoe,
- \$1140 for Pacific,
- \$800 for Savannah, and
- \$400 for Aspen.

The factory operates under limited resource availability. During the planning period, there are 5,800 quarts of glue available, 780 pressing machine hours, and 29,200 pounds of pine chips. In addition, the supply of oak chips is limited to a fixed total amount.

Table 1: Resources Required per Pallet (50 units) of Panelling Type

	Tahoe	Pacific	Savannah	Aspen
Glue (quarts)	50	50	100	50
Pressing (hours)	5	15	10	5
Pine chips (pounds)	500	400	300	200
Oak chips (pounds)	500	750	250	500

Let decision variables be X_1, X_2, X_3, X_4 (pallets of Tahoe, Pacific, Savannah, Aspen).

Objective

Constraints

Example 3: Diet optimisation

Foods: Brownies (x_0), Ice Cream (x_1), Cola (x_2), Cheesecake (x_3). Nutrition values and cost per unit:

Table 2: Nutrition values and cost per unit

	Brownies	Ice Cream	Cola	Cheesecake
Calories	400	200	150	500
Chocolate	3	2	0	0
Sugar	2	2	4	4
Fat	2	4	1	5
Cost ($)	0.50	0.20	0.30	0.80

Goal: **minimum-cost** diet with at least 500 calories, 6g chocolate, 10g sugar, 8g fat.

Objective

Constraints

Example 4: Transportation (TowAlong trailers)

Plants: Kansas City, Denver, Raleigh. Distribution centres: Birmingham, Milwaukee, Los Angeles, Seattle. Costs, capacities, and demands are given below.

Table 3: Unit shipping costs, plant capacities, and distribution demands

Plant	Birmingham	Milwaukee	Los Angeles	Seattle	**Capacity**
Kansas City	35	40	65	120	12000
Denver	30	30	45	130	8000
Raleigh	60	65	50	100	5000
Demand	9000	3000	9500	1500	

Let x_{ij} be shipments from plant i to distribution centre j.

Objective

Constraints

subject to capacity constraints

and demand constraints

Exercise

1. Sunchem ink production and distribution

Sunchem, a manufacturer of printing inks, has five manufacturing plants worldwide. Their locations and capacities are shown in Table 4 along with the cost of producing 1 ton of ink at each facility. The production costs are in the local currency of the country where the plant is located. The major markets for the inks are North America, South America, Europe, Japan, and the rest of Asia. Demand at each market is shown in Table 4. Transportation costs from each plant to each market in U.S. dollars are shown in Table 4. Management must come up with a production plan for the next year.

a) If exchange rates are expected as in Table 5, and no plant can run below 50 percent of capacity, how much should each plant produce and which markets should each plant supply?

b) If there are no limits on the amount produced in a plant, how much should each plant produce?

c) Can adding 10 percent of capacity in any plant reduce costs?

d) How should Sunchem account for the fact that exchange rates fluctuate over time?

Table 4: Transportation Costs, Production Capacity, and Cost by Region

Supply Region	**N. America**	**S. America**	**Europe**	**Japan**	**Asia**	**Capacity (tons/year)**	**Production Cost per ton**
United States	600	1,200	1,300	2,000	1,700	185	\$10,000
Germany	1,300	1,400	600	1,400	1,300	475	€15,000
Japan	2,000	2,100	1,400	300	900	50	¥1,800,000
Brazil	1,200	800	1,400	2,100	2,100	210	13,000 real
India	2,200	2,300	1,300	1,000	800	80	400,000
Demand (tons/year)	270	190	200	120	100		

Table 5: Exchange Rates as of April 22, 2025

	USD	EUR	JPY	BRL	INR
USD	1.0000	0.8794	141.94	5.6336	85.282
EUR	1.1374	1.0000	161.45	6.408	97.0
JPY	0.0070	0.0062	1.0000	0.045	0.60
BRL	0.1775	0.1560	22.22	1.0000	15.14
INR	0.0117	0.0103	1.67	0.066	1.0000

Sunchem has $n = 5$ manufacturing plants and $m = 5$ regional markets.

Notation:

- D_j: annual tons needed in market j
- K_i: maximum capacity of plant i
- For part (a): lower limit is $0.5K_i$
- c_{ij}: shipping cost per ton from plant i to market j (USD)
- p_i: production cost per ton at plant i (local currency)
- x_{ij}: tons shipped from plant i to market j (integer, nonnegative)

Model statement (minimise overall network cost)

A network-cost formulation is

subject to

2. Home office locations (fixed + trip costs)

A supply chain consulting firm must decide on the location of its home offices. Its clients are located primarily in the 16 states listed in the Table below. There are four potential sites for home offices: Los Angeles, Tulsa, Denver, and Seattle. The annual fixed costs of locating an office in Los Angeles, Tulsa, Denver, and Seattle are $165,428, $131,230, $140,000, and $145,000, respectively. The expected number of trips to each state and the travel costs from each potential site are shown in Table 6. Each consultant is expected to take no more than 25 trips per year.

a) If there are no restrictions on the number of consultants at a site and the goal is to minimize costs, where should the home offices be located and how many consultants should be assigned to each office? What are the annual costs associated with the facility and travel?

b) If at most 10 consultants are to be assigned to a home office, where should the offices be located? How many consultants should be assigned to each office? What is the annual cost of this network?

c) What do you think of a rule that assigns all consulting projects from a given state to a single home office? How much is this policy likely to add to the cost compared to allowing multiple offices to handle a single state?

Table 6: The expected number of trips to each state and the travel costs from each potential site

State	L.A.	Tulsa	Denver	Seattle	Trips
Washington	150	250	200	25	40
Oregon	150	250	200	75	35
California	75	200	150	125	100
Idaho	150	200	125	125	25
Nevada	100	200	125	150	40
Montana	175	175	125	125	30
Wyoming	150	175	100	150	50
Utah	150	150	100	200	30
Arizona	75	150	100	250	30
Colorado	150	125	25	250	65
New Mexico	125	125	75	300	40
North Dakota	300	200	150	200	30
South Dakota	300	175	125	200	20
Nebraska	250	100	125	250	30
Kansas	250	75	75	300	30
Oklahoma	250	25	125	300	55

Overall cost combines fixed costs of opening home offices plus trip costs.

Notation:

- $n = 4$ potential home office locations, $m = 16$ states
- D_j annual trips needed to state j
- f_i annual fixed cost of opening office i
- c_{ij} trip cost from office i to state j
- $y_i \in \{0, 1\}$ equals 1 if office i is open
- x_{ij} number of trips from office i to state j (integer, nonnegative)

(a)

Objective

Typical constraints include

(b)

Objective

Typical constraints include

(c)

Objective

Typical constraints include

3. Dry Ice, Inc.

Dry Ice, Inc., a manufacturer of air conditioners, has seen its demand grow significantly. The company anticipates nationwide demand for 2025 to be 180,000 units in the South, 120,000 units in the Midwest, 110,000 units in the East, and 100,000 units in the West. Managers at Dry Ice are designing the manufacturing network and have selected four potential sites: New York, Atlanta, Chicago, and San Diego. Plants could have a capacity of either 200,000 or 400,000 units. The annual fixed costs at the four locations are shown in Table 4, along with the cost of producing and shipping an air conditioner to each of the four markets. Where should Dry Ice build its factories, and how large should they be?

Table 7: Production and Transportation Costs for Dry Ice, Inc.

	New York	**Atlanta**	**Chicago**	**San Diego**
Annual fixed cost of 200,000 plants	\$6 million	\$5.5 million	\$5.6 million	\$6.1 million
Annual fixed cost of 400,000 plants	\$10 million	\$9.2 million	\$9.3 million	\$10.2 million
East	\$211	\$232	\$238	\$299
South	\$232	\$212	\$230	\$280
Midwest	\$240	\$230	\$215	\$270
West	\$300	\$280	\$270	\$225

Notation:

- $n = 4$ potential sites, $m = 4$ regional markets
- D_j annual demand in market j
- K_i capacity of site i (each $K_i = 400{,}000$)
- If needed capacity $\leq 200{,}000$, use the corresponding fixed cost; otherwise use the $400{,}000$ fixed cost
- f_i fixed cost of opening site i
- c_{ij} cost of producing and shipping one unit from site i to market j
- $y_i \in \{0, 1\}$ open/close decision
- $x_{ij} \in \mathbb{Z}_{\geq 0}$ shipments from i to j

Objective

subject to

4. BAKCELL and DCELL merger (network design)

BAKCELL and DCELL are two manufacturers of cell phones that have recently merged. Their current market sizes are shown in Table 8. All demand is in millions of units.

BAKCELL has three production facilities in Europe (EU), North America, and South America. DCELL also has three production facilities in Europe (EU), North America, and Rest of Asia. The capacity (in millions of units), annual fixed cost (in millions of $), and variable production costs ($ per unit) for each plant are as shown in Table 9. Transportation costs between regions are as shown in Table 10. All transportation costs are shown in $ per unit.

Duties are applied to each unit based on the fixed cost per unit capacity, variable cost per unit, and transportation cost. Thus, a unit currently shipped from North America to Africa has a fixed cost per unit of capacity of $5.00, a variable production cost of $5.50, and a transportation cost of $2.20. The 25% import duty is thus applied on $12.70 (5.00 + 5.50 + 2.20) to give a total cost on import of $15.90. For the questions below, assume that market demand is in Table 5.

The merged company has estimated that scaling back a 20-million-unit plant to 10 million units saves 30% in fixed costs. Variable costs at a scaled-back plant are unaffected. Shutting a plant down (10 million or 20 million units) saves 80 % in fixed costs. Fixed costs are only partially recovered because of severance and other costs associated with a shutdown.

a) What is the lowest cost achievable for the production and distribution network before the merger? Which plants serve which markets?

b) If none of the plants is shut down, what is the lowest cost achievable for the production and distribution network after the merger? Which plants serve which markets?

c) If plants can be scaled back or shut down in batches of 10 million units of capacity after the merger, what is the lowest cost achievable for the production and distribution network? Which plants serve which markets?

d) How is the optimal network configuration affected if all duties are reduced to 0?

e) How should the merged network be configured?

Duty calculation

Duties apply to (fixed-cost per unit capacity) + (variable production cost) + (transportation cost). Example given: shipping from North America to Africa has fixed $5.00, variable $5.50, transport $2.20. A 25% duty applies to $12.70 producing total import cost $15.90.

Table 8: Global demand and duties for BAKCELL and DCELL

Market	North America	South America	Europe (EU)	Europe (non-EU)	Japan	Rest of Asia/Australia	Africa
BAKCELL demand	10	4	20	3	2	2	1
DCELL demand	12	1	4	8	7	3	1
Import duties (%)	3	20	4	15	4	22	25

Table 9: Plant capacities and costs for BAKCELL and DCELL

		Capacities	Fixed Cost/Year	Variable Cost/unit
3***BAKCELL**	Europe (EU)	20	100	6.0
	N. America	20	100	5.5
	S. America	10	60	5.3
3***DCELL**	Europe (EU)	20	100	6.0
	N. America	20	100	5.5
	Rest of Asia	10	50	5.0

Table 10: Transportation costs between Regions ($ per unit)

	North America	South America	Europe (EU)	Europe (non EU)	Japan	Rest of Asia/Australia	Africa
N. America	1.00	1.50	1.50	1.80	1.70	2.00	2.20
S. America	1.50	1.00	1.70	2.00	1.90	2.20	2.20
Eu	1.50	1.70	1.00	1.20	1.80	1.70	1.40
non EU	1.80	2.00	1.20	1.00	1.80	1.60	1.50
Japan	1.70	1.90	1.80	1.80	1.00	1.20	1.90
Asia/Australia	2.00	2.20	1.70	1.60	1.20	1.00	1.80
Africa	2.20	2.20	1.40	1.50	1.90	1.80	1.00

Notation

- $\mathcal{I} = \{1, \ldots, 6\}$: index set of the 6 plants (e.g., BAK-EU, BAK-NA, BAK-SA, DCE-EU, DCE-NA, DCE-Asia).
- $\mathcal{J} = \{1, \ldots, 7\}$: index set of the 7 markets/regions.
- D_j: demand (in millions of units) for region j.
- K_i: capacity (in millions of units) of plant i.
- F_i: annual fixed cost (in $ millions) of operating plant i at full capacity.
- c_{ij}: *per-unit* cost (in $) of producing at plant i and delivering to region j.

- $x_{ij} \geq 0$: decision variable = number of units (in millions) produced at plant i and shipped to region j.

Questions

1. Lowest cost achievable for the production/distribution network *before* the merger; which plants serve which markets?
2. If none of the plants is shut down, lowest cost *after* the merger; which plants serve which markets?
3. If plants can be scaled back or shut down in batches of 10 million units of capacity after the merger, lowest cost and assignments.
4. Effect on optimal network configuration if all duties are reduced to 0.
5. How should the merged network be configured?